

1. Kubernetes scalability

Yes. The official Kubernetes v1.37 documentation explicitly states that Kubernetes is designed to accommodate clusters meeting all of these limits:

- **Maximum 5,000 nodes**
- **Maximum 150,000 total Pods**
- **Maximum 300,000 total containers**
- **Maximum 110 Pods per node**

These figures appear together in Kubernetes' official “**Considerations for large clusters**” documentation. ([Kubernetes](#))

Verification result: Confirmed.

2. Docker Swarm manager limits

Yes. Official Docker documentation states that Docker recommends a **maximum of seven manager nodes** for a swarm.

The same documentation gives the Raft fault-tolerance relationships:

- **3 managers** → **tolerates loss of 1 manager**
- **5 managers** → **tolerates simultaneous loss of 2 managers**
- **7 managers** → **tolerates loss of 3 managers**

More generally, Docker states that an odd number **N** of managers tolerates loss of at most **(N-1)/2** managers. ([Docker Documentation](#))

Therefore, the stated 3/5/7 relationship follows directly from the documented Raft rule.

Verification result: Confirmed.

3. AWS Fargate resource limits

Yes, with an important platform/OS qualification.

AWS's official ECS documentation lists these valid Fargate task-level CPU/memory combinations: ([AWS Documentation](#))

CPU	Supported memory	OS / platform qualification
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0.25 vCPU	0.5, 1, 2 GB	Linux
0.5 vCPU	1–4 GB	Linux
1 vCPU	2–8 GB	Linux, Windows
2 vCPU	4–16 GB, 1-GB increments	Linux, Windows
4 vCPU	8–30 GB, 1-GB increments	Linux, Windows
8 vCPU	16–60 GB, 4-GB increments	Linux, platform 1.4.0+
16 vCPU	32–120 GB, 8-GB increments	Linux, platform 1.4.0+
32 vCPU	60, 120, or 244 GB	Linux, platform 1.4.0+

Thus:

- **0.25 vCPU with 0.5–2 GB: Supported**, specifically 0.5 GB, 1 GB, or 2 GB on Linux.
- **32 vCPU with up to 244 GB: Supported**, with the documented memory choices being **60 GB, 120 GB, and 244 GB**, on Linux platform 1.4.0 or later.

So the shorthand “**0.25 vCPU with 0.5–2 GB**” and “**32 vCPU with up to 244 GB**” is substantively correct, but the latter should not imply every memory value between 60 and 244 GB is valid—the documented choices are discrete. ([AWS Documentation](#))

Verification result: Confirmed, with the discrete-memory/platform qualification above.

4. Fargate pricing

Yes. AWS's official Fargate pricing page explicitly gives these rates in its **US East (N. Virginia), Linux/X86** pricing example:

- **vCPU: \$0.000011244 per vCPU-second**
- **Memory: \$0.000001235 per GB-second**
- **Ephemeral storage: \$0.0000000308 per GB-second** for additional ephemeral storage ([Amazon Web Services, Inc.](#))

The pricing page also specifies that **20 GB of ephemeral storage is provided by default** for Fargate tasks and Pods; customers pay for **additional storage configured beyond that amount**. Its example uses 30 GB total storage, so only **10 GB** is charged. ([Amazon Web Services, Inc.](#))

For current ECS Fargate task storage, AWS documents:

- **Default:** 20 GiB
- **Maximum:** 200 GiB

- Additional configurable storage requires Linux platform **1.4.0+** or Windows platform **1.0.0+**. ([AWS Documentation](#))

The AWS pricing page's US East example therefore verifies the exact three rates quoted above. ([Amazon Web Services, Inc.](#))

Verification result: Confirmed.

Overall audit result

Question	Result	Key qualification
Kubernetes 5,000 / 150,000 / 300,000 / 110	Confirmed	All four figures are explicitly documented
Swarm ≤ 7 managers; 3/5/7 failure tolerance	Confirmed	Based directly on Docker's Raft rule
Fargate 0.25 vCPU $\rightarrow$ 0.5–2 GB	Confirmed	Linux; three discrete memory values
Fargate 32 vCPU $\rightarrow$ up to 244 GB	Confirmed	Linux platform 1.4.0+; 60/120/244 GB choices
Fargate US-East Linux/x86 rates	Confirmed	\$0.000011244/vCPU-s; \$0.000001235/GB-s
Additional ephemeral storage	Confirmed	\$0.0000000308/GB-s beyond the default 20 GB

These results are based on the cited **official Kubernetes, Docker, and AWS documentation**, rather than third-party summaries.